Divyam Singh Duhoon

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Education

National Institute of Technology, Hamirpur **2024 – 2028**

Bachelor of Technology in Computer Science and Engineering

JEE Main 2024: 98.9 percentile among 1.4M+ unique candidates

Work Experience

AI/ML Research Intern — Certificate **May 2026 – July 2026**

Indian Institute of Technology (BHU), Varanasi *PyTorch, Transformers, Deep Learning, Computer Vision, Multimodal AI*

- Reproduced and fine-tuned **Uni-Sign** using **PyTorch** and **Transformers** for **Sign Language Understanding**, gaining hands-on experience in deep learning and multimodal AI.
- Developed a **3-modal AI pipeline** integrating pose, visual, and textual representations for sign language translation and sequence understanding.
- Applied **transfer learning** to adapt Chinese Sign Language models to Indian Sign Language datasets, evaluating model adaptation across different domains.

Frontend Developer Intern — Certificate **September 2025 – May 2026**

National Institute of Technology, Hamirpur *Next.js, TypeScript, Tailwind CSS, REST APIs*

- Developed a production-scale government educational **web application** using **Next.js, TypeScript, and Tailwind CSS**.
- Built modular frontend components using **Next.js App Router** and collaborated with faculty, designers, and backend teams to deliver application features.
- Integrated frontend workflows with **backend services and REST APIs**, contributing to end-to-end application development.
- Improved page-load performance by **30%** through layout optimization, asset compression, and framework-level optimizations.

Skills

- **Languages:** Python, C++, JavaScript, TypeScript
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Problem Solving
- **AI/ML:** Machine Learning, Deep Learning, Neural Networks, Multimodal AI, Computer Vision, Transfer Learning, PyTorch, TensorFlow, Transformers, OpenCV, MediaPipe
- **Web Development:** React.js, Next.js, Node.js, Express.js, REST APIs, HTML, CSS, Tailwind CSS
- **Databases:** SQL, MongoDB, Mongoose
- **Tools & Platforms:** Git, GitHub, Linux, VS Code, Chrome DevTools, Figma

Projects

BridgeTheGap [Developed during Electrothon 7.0] **Python, ReactJS, OpenCV, MediaPipe**

- Developed an **AI-powered real-time Sign Language-to-Speech** prototype using **Python, OpenCV, and MediaPipe** for accessibility-focused communication.
- Built a **computer vision and gesture recognition pipeline** tracking **21-point hand landmarks** for real-time sign interpretation.
- Optimized gesture recognition inference for **real-time responsiveness** and improved robustness across diverse hand poses and input conditions.

HimadriAllInOne: Hostel Management Platform **MERN, REST APIs**

- Architected a modular **full-stack MERN application** integrating mess attendance, meal availability, billing summaries, and student activity tracking.
- Developed **React, Node.js, Express.js, and MongoDB** workflows with REST APIs to support **300+ users** across hostel management operations.
- Designed scalable dashboards and optimized **data workflows** for centralized hostel operations and improved application efficiency.
- Consolidated multiple hostel utilities into a single platform, improving administrative efficiency by **35%**.

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Leadership & Volunteering Experience

- Serve as **Domain Lead, SHEILD, NITH**, driving cybersecurity initiatives and advancing hands-on security awareness and technical capability across the student community.
- Serve as **Web Coordinator, E-Cell NITH**, overseeing web development initiatives and enabling digital solutions that support entrepreneurial programs and institutional outreach.
- Promoted to **Core Coordinator, Team Technical, NITH**, spearheading technical initiatives and aligning cross-functional teams to execute institute-level activities.
- Led a **6-member multidisciplinary team** representing **NIT Hamirpur** at **Smart India Hackathon (SIH) 2025**, driving collaborative problem-solving and end-to-end solution development.
- Coordinated participant management, event logistics, and technical operations for **Hack 5.0**, organized by **CSEC, NIT Hamirpur**, ensuring seamless execution of event activities.